

Homework 1

Drude Model

1. (10pts) Describe (briefly) the four key assumptions of the Drude Model.
2. (15pts) Consider the metallic elements Ag, Cu, Mg and Ca.
 - (a) Find the mass densities in g/cm³ units.
 - (b) Calculate the respective electron densities units of electrons per cm³.
 - (c) Use the electrical resistivities in Table 1.2 in Ashcroft and Mermin to obtain the relaxation times at 77K and 273K.
3. (15pts) Calculate the plasma frequencies for Au and Ag. Will the surfaces of these two metals display identical mirror-like properties for light in the visible range of the spectrum (8000 - 4000 Å) ? Is your answer consistent with the colors of Au and Ag surfaces?
4. (30pts) Here we will explore the frequency dependent magnetoconductivity tensor within the Drude model. Assume a metal with a density of n free electrons in a static magnetic field H along the z -direction. The electric current density in the xy plane is related to the electric field by

$$J_x = \sigma_{xx}E_x + \sigma_{xy}E_y \qquad J_y = \sigma_{yy}E_y + \sigma_{yx}E_x$$

Assume that the frequency $\omega \gg \omega_c$ and $\omega \gg 1/\tau$, where $\omega_c = \frac{eH}{mc}$ and τ is the scattering time. Solve the equations of motion:

$$\begin{aligned} \frac{dp_x}{dt} + \frac{p_x}{\tau} &= -eE_x - \omega_c p_y \\ \frac{dp_y}{dt} + \frac{p_y}{\tau} &= -eE_y + \omega_c p_x \\ \frac{dp_z}{dt} + \frac{p_z}{\tau} &= -eE_z \end{aligned}$$

to find the components of the magneto-conductivity tensor:

$$\sigma_{xx} = \sigma_{yy} = \frac{i\omega_p^2}{4\pi\omega} \qquad \sigma_{xy} = -\sigma_{yx} = \frac{\omega_c\omega_p^2}{4\pi\omega^2}$$

where $\omega_p^2 = \frac{4\pi ne^2}{m}$.

5. (30 pts) Here we will explore the *static* magnetoconductivity tensor within the Drude model. Using the equations of motion from the above problem, show that the static current density can be written in matrix form as

$$\begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix} = \frac{\sigma_o}{1 + (\omega_c \tau)^2} \begin{pmatrix} 1 & -\omega_c \tau & 0 \\ \omega_c \tau & 1 & 0 \\ 0 & 0 & 1 + (\omega_c \tau)^2 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}$$

In the high magnetic field limit $\omega_c \tau \gg 1$, show that

$$\sigma_{yx} = \frac{ne c}{H} = -\sigma_{xy}$$

In this limit $\sigma_{xx} = 0$, to order $\frac{1}{\omega_c \tau}$. The Hall conductivity is σ_{xy} .